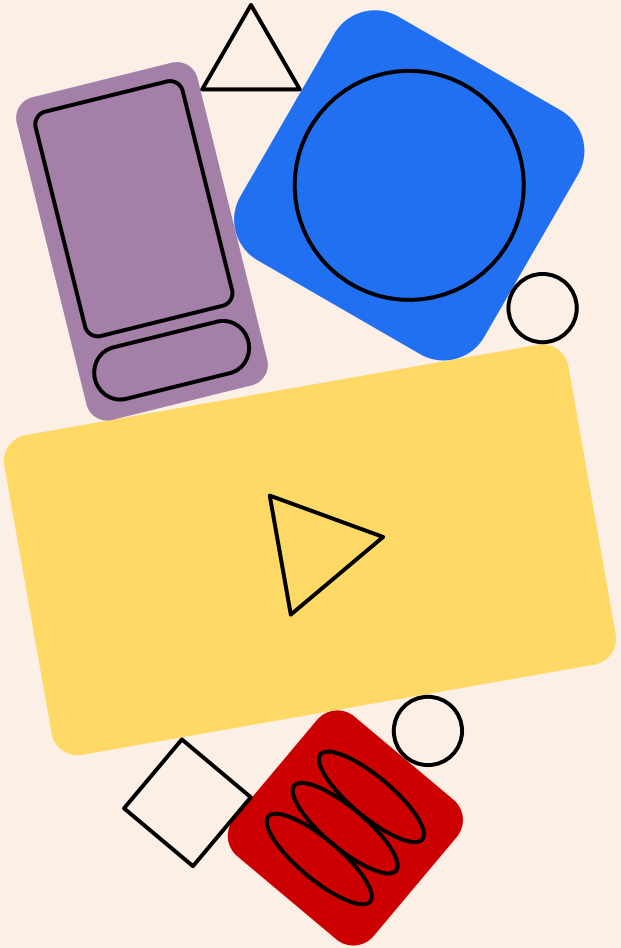
ISM 102 - Problem Solving Techniques using Computers

Topic 2: Systematic Problem Solving & Algorithms



Recap

Let's recall..

- What is a problem?
- Difference between well-defined & ill-defined?
- What is an algorithm?

Why Do We Need a Systematic Approach?

Can we directly start coding?

Most students/developers make this mistake:

- Jump to code → Get confused → Debug forever

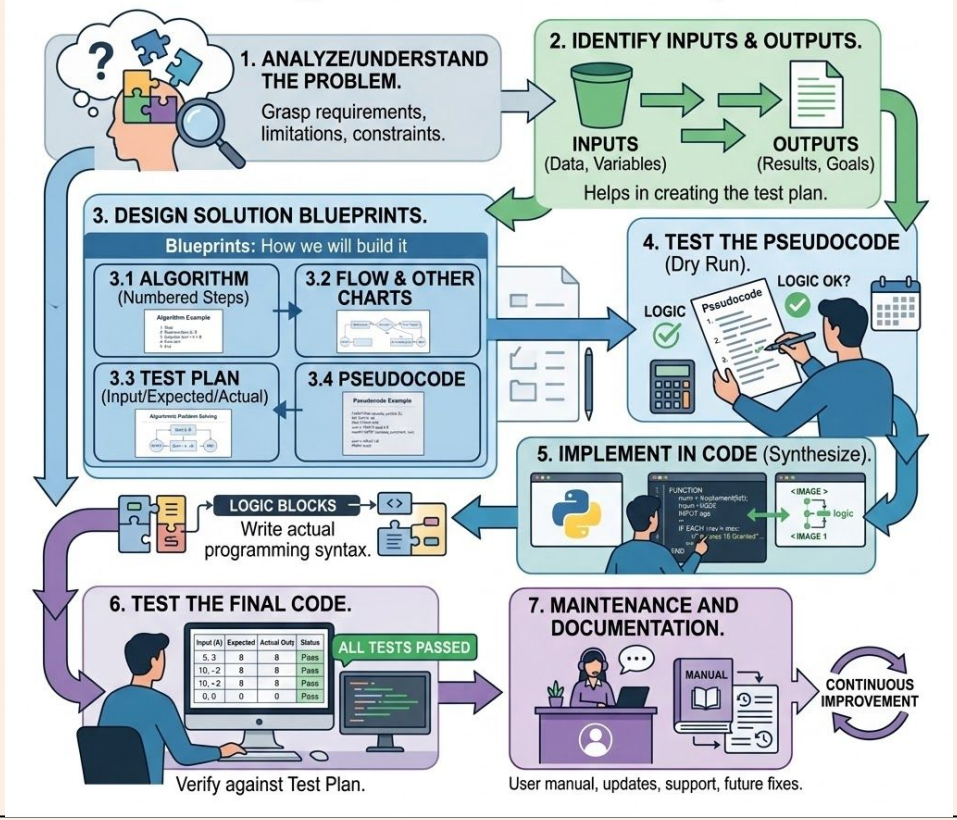
The correct approach is:

- Think → Design → Then code

What does design include?

- Algorithm
- Flowchart and other charts
- Test plan
- Pseudocode

The Steps in Problem Solving



The Real Thinking happens before coding!

Understand the Problem

- What is required?
- What is given?
- What are the constraints/rules?

Example: “Calculate a student’s final grade”

Ask Questions:

- How many assessments?
- Weightage?
- Passing marks?

Identify Input & Output

Example: Calculate Student Exam Percentage. A student scored marks in 5 subjects. Each subject is out of 100.

Inputs:

- Marks for Subject1
- Marks for Subject2
- Marks for Subject3
- Marks for Subject4
- Marks for Subject5

(Validation: Each mark must be between 0 and 100)

Output:

- Total marks
- Percentage

Design the Algorithm

Rules: Clear, Finite, Logical & Ordered

Example: Find the Average of Three Numbers

Start

- Get three numbers as input (let's call them A, B, and C).
- **Check if the three numbers are valid. If invalid stop or ask again.**
- Add the three numbers together and call it Sum (Sum = A + B + C).
- Divide the Sum by 3 to get the Result (Result = Sum / 3).
- Display the Result.

End

Example

Problem: Calculate monthly savings.

A Student earns 2000 AED allowance. Spends 1200 AED. Find savings!

Inputs & Outputs:

Income = input

Expenses = input

Savings = output

Constraints:

Income must be ≥ 0

Expenses must be ≥ 0

Expenses cannot exceed income

Algorithm:

Start

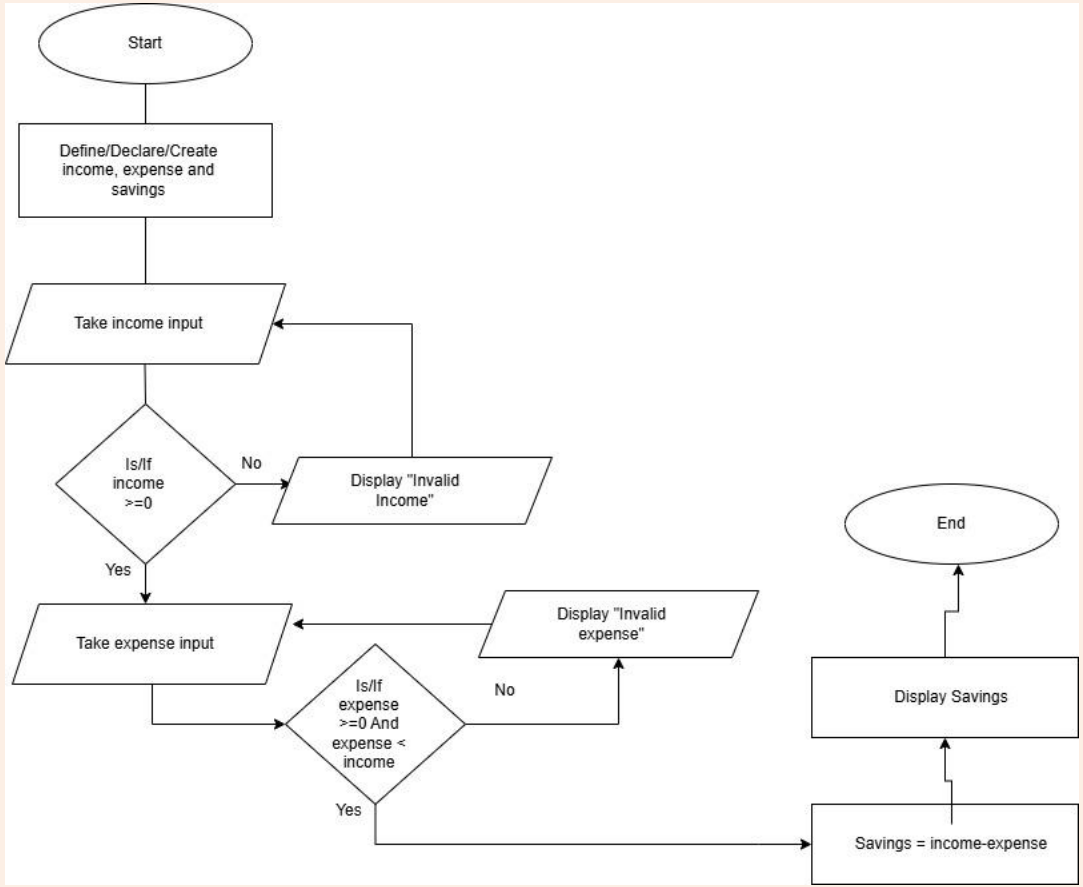
- Take input of income
  - If income < 0 Display "Invalid income" and Stop or ask again.
- Take input of expenses
  - If expenses < 0 OR expenses > income Display "Invalid expenses" and Stop or ask again.
- Deduct expenses from income to find savings.
- Display savings

Stop

Example

Flowchart

You can use draw.io to create the flowchart diagram.



Example

Pseudocode

```
Start
        //Declaring income and expense variables
        DECLARE income, expense : REAL
        //Declaring and initializing savings variable
        DECLARE savings = 0
        DISPLAY “Enter your income”
        INPUT income
        IF income < 0
                Display "Invalid income"
                Go to End/Go back to previous step
        END IF
        OUTPUT “Enter your expense”
        INPUT expense
        IF expense < 0 AND expense > income
                Display "Invalid expense"
                Go to End/Go back to previous step
        END IF
        //Assigning result to savings variable
        Savings = income - expense
        PRINT “Your savings are ” + savings
End
```

Example

Testing

Test No.	Test Case	Inputs	Expected Output
1	Valid Input	1000,500	500
2		1250, 450.5	799.5
3	Invalid Input	0,200	Invalid Expense
4		-2000	Invalid Income
5		2000, -400	Invalid Expense
6		5000, ABC	Invalid Expense
7		XYZ	Invalid Income
8	Savings Calculation	2250, 650	1600

Example

Code - Implementation in Python (Just Introduction - NO NEED TO LEARN NOW)

```
income = float(input("Enter your monthly income: "))
if income < 0:
    print("Invalid income entered.")
else:
    expenses = float(input("Enter your monthly expenses: "))
    if expenses < 0 or expenses > income:
        print("Invalid expenses entered.")
    else:
        savings = income - expenses
        print("Your monthly savings are:", savings)
```

Class Activity

Problem: Calculate Final Movie Ticket Cost. A cinema charges: 35 AED per ticket. If the customer buys 5 or more tickets, they get a 10% discount on total amount.

Requirements:

Ask user to enter number of tickets

Validate:

Number of tickets must be greater than 0

Calculate total cost

Apply discount if applicable

Display final amount

What to do?

Follow steps of problem solving from 1 to 4 for this problem and document your solution.

Submission Deadline - 10th March 2026 until 11:59 pm.

Thank you!